FBISE / FEDERAL BOARD • HSSC-I / 1ST YEAR

Physics
Class 11

156 Three-Mark Theory Questions & Answers

Summative Topics • Exam Preparation

Complete questions followed by three point-wise answers, in simple English.

Chapter priority order

9, 8, 12, 11, 6, 4, 14, 13, 5, 3, 2, 10, 1, 7

The questions, answers and equations in this booklet reproduce the complete 156-question bank prepared in the chat. No answer points have been shortened or omitted.

These are exam-style practice questions selected using the uploaded summative assessment framework and relevant textbook content. Each question is intended as a three-mark response; the actual division of marks depends on the examiner’s marking scheme.

Made by Fakhar STEM Sphere
fakharstemsphere.com

Chapter index

Questions follow the requested priority order. Select a chapter to jump to its first page.

Chapter 9 - Waves [Q1-23] . . . 3

Chapter 8 - Heat and Thermodynamics [Q24-40] . . . 7

Chapter 12 - Magnetism / Electromagnetism [Q41-55] . . . 10

Chapter 11 - Electricity [Q56-69] . . . 13

Chapter 6 - Fluid Mechanics [Q70-81] . . . 16

Chapter 4 - Rotational and Circular Motion [Q82-94] . . . 18

Chapter 14 - Particle Physics [Q95-106] . . . 21

Chapter 13 - Relativity [Q107-115] . . . 23

Chapter 5 - Work and Kinetic Energy [Q116-123] . . . 25

Chapter 3 - Translatory Motion [Q124-131] . . . 27

Chapter 2 - Vectors [Q132-138] . . . 29

Chapter 10 - Electrostatics [Q139-145] . . . 31

Chapter 1 - Physical Quantities and Measurements [Q146-150] . . . 33

Chapter 7 - Physics of Solids [Q151-156] . . . 34

Source documents

FBISE Physics Book Class 11 Federal Board.pdf
FBISE_Physics_Class11_Summative_Formative_Topics(1).pdf

Theory short questions only. No MCQ sets, calculation exercises or long derivations are included.

Chapter 9 - Waves

Questions 1-23 | 23 questions | 3 marks each

1. Define the intensity of a progressive wave. How does it depend on amplitude?

- Intensity is the wave energy passing per second through unit area perpendicular to the direction of propagation.
- It is given by $I = P/A$, where P is power and A is area. Its SI unit is $\mathrm{W\,m^{-2}}$.
- Under the same conditions, intensity is proportional to the square of displacement amplitude: $I \propto a^2$.

2. What is the Doppler effect? Explain it for a moving source and a stationary observer.

- The Doppler effect is the apparent change in observed frequency caused by relative motion between a source and an observer.
- When the source approaches a stationary observer, the observed frequency is higher than the emitted frequency.
- When the source moves away, the observed frequency is lower than the emitted frequency.

3. Why does the pitch of an ambulance siren decrease as it passes a stationary listener?

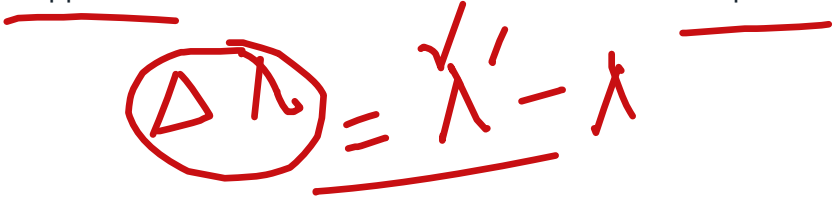
- During approach, successive wavefronts reach the listener closer together, producing a shorter wavelength.
- Since the speed of sound in the medium is approximately unchanged, the shorter wavelength produces a higher observed frequency.
- After the ambulance passes, wavefronts reaching the listener are farther apart, so the observed frequency and pitch decrease.

4. How is the Doppler effect used in studying cardiac problems?

- Doppler ultrasound sends high-frequency sound waves into the heart and blood vessels.
- Moving blood cells reflect these waves with a changed frequency; the frequency shift provides information about blood-flow speed and direction.
- Abnormal flow patterns help doctors identify narrowed arteries, obstructed blood flow, and leaking or narrowed heart valves.

5. Explain how a radar speed trap measures the speed of a vehicle.

- The radar sends radio waves of a known frequency towards a moving vehicle.
- The reflected waves have a different frequency because of the vehicle's motion.
- The instrument measures this Doppler shift and uses it to determine the vehicle's speed along the radar beam.



6. How does SONAR use the Doppler effect to study a moving submarine?

- SONAR transmits sound waves through water towards the submarine.
- The moving submarine reflects sound waves with a changed frequency.
- The frequency shift provides information about its speed towards or away from the SONAR system.

7. Explain the use of the Doppler effect in astronomy.

- Astronomers compare the observed frequencies of a star's spectral lines with their known laboratory frequencies.
- A shift towards higher frequency, called a blue shift, indicates motion towards the observer.
- A Doppler shift towards lower frequency, called a red shift, indicates motion away; its size gives information about radial speed.

8. How can the Doppler effect help determine the motion of an artificial satellite?

- An artificial satellite transmits radio signals of a known frequency.
- A receiving station detects a frequency shift as the satellite moves towards or away from it.
- Analysis of this shift gives information about the satellite's velocity along the line joining it to the station.

9. What is polarization? Why does it establish that light is transverse?

- Polarization is the restriction of a transverse wave's vibrations to one direction perpendicular to propagation.
- Light can be polarized, showing that its electric-field vibrations occur perpendicular to its direction of travel.
- Longitudinal waves cannot be polarized because their vibrations are already along the direction of propagation.

10. State Malus's law and explain the effect of rotating an analyser.

- Malus's law states that the transmitted intensity is $I = I_0 \cos^2 \theta$, where I_0 is the incident plane-polarized intensity.
- Here, θ is the angle between the incident light's polarization direction and the analyser's transmission axis.
- Transmission is maximum for parallel directions and ideally zero for perpendicular directions.

11. How are Polaroids used in sky photography?

- Light scattered by the atmosphere is partly polarized and can make the sky appear bright.
- A suitably oriented Polaroid filter reduces part of this polarized background light.
- The darker sky improves contrast, making clouds and other features appear clearer.

12. Explain how Polaroids are used in stress analysis of transparent materials.

- A transparent specimen is placed between crossed polarizing filters and illuminated.
- Stress changes the specimen's optical properties, producing a pattern of coloured bands.
- The pattern reveals the distribution of stress; closely concentrated bands indicate regions of greater stress concentration.

13. State the principle of superposition of waves and explain its meaning.

- When waves overlap, the resultant displacement at a point is the vector sum of their individual displacements.
- For displacements along the same line, this becomes the algebraic sum: $y = y_1 + y_2 + \cdots$.
- In a linear medium, the waves pass through one another and continue travelling after the overlap.

14. Distinguish between constructive and destructive interference.

- Constructive interference occurs when waves meet in phase; destructive interference occurs when they meet in opposite phase.
- In constructive interference, displacements reinforce; in destructive interference, they oppose one another.
- Constructive interference produces maximum intensity, while destructive interference produces minimum intensity, ideally zero for equal amplitudes.

15. How do noise-cancelling headphones use the principle of superposition?

- A microphone detects unwanted sound from the surroundings.
- The headphones produce a sound wave of approximately equal amplitude and opposite phase.
- The two waves superpose destructively, reducing the unwanted sound reaching the ear.

16. Explain the formation of stationary waves.

- Stationary waves form when two waves of equal frequency and amplitude travel in opposite directions through the same medium.
- Their superposition produces fixed nodes of zero displacement and antinodes of maximum displacement.
- The pattern remains stationary and has no net energy transfer along the medium.

17. Explain the formation of harmonics on a string fixed at both ends.

- Both fixed ends must be displacement nodes, so only particular wavelengths can form stationary patterns.
- The fundamental pattern contains one loop, with a node at each end and an antinode at the centre.
- Higher harmonics contain more loops, and their frequencies are integer multiples of the fundamental: $f_n = nf_1$.

18. Why does an open pipe produce all harmonics, whereas a pipe closed at one end produces only odd harmonics?

- An open end is approximately a displacement antinode, while a closed end is a displacement node.
- An open pipe accommodates an integer number of half-wavelengths, producing all integer harmonics.
- A pipe closed at one end accommodates odd numbers of quarter-wavelengths, producing only odd harmonics.

19. Explain diffraction of water waves through a narrow gap.

- Diffraction is the spreading of waves as they pass through a gap or around an obstacle.
- In a ripple tank, plane water waves passing through a gap comparable to their wavelength spread into curved wavefronts.
- Spreading becomes less pronounced when the gap is much wider than the wavelength.

20. What are coherent sources? Why are they required for a stable interference pattern?

- Coherent sources emit waves of the same frequency with a constant phase difference.
- Their constant phase relationship keeps positions of constructive and destructive interference fixed.
- Incoherent sources have a changing phase relationship, so a stable interference pattern is not maintained.

21. What are beats? Explain their production.

- Beats are periodic increases and decreases in loudness produced by overlapping sound waves of slightly different frequencies.
- The waves alternately reinforce and oppose one another because their relative phase changes continuously.
- The beat frequency is the magnitude of their frequency difference: $f_b = |f_1 - f_2|$.

22. Explain the use of beats in tuning musical instruments.

- The instrument is sounded together with a standard tuning fork of known frequency.
- Beats indicate that the two frequencies are slightly different; the instrument’s tension or effective length is adjusted.
- As the frequencies become equal, the beat frequency approaches zero and the beats disappear.

23. What are gravitational waves? Describe their source and propagation.

- Gravitational waves are travelling disturbances, or ripples, in spacetime.
- Massive bodies accelerating in an orbiting binary system can produce gravitational waves.
- These waves carry energy away from their source and travel through vacuum at the speed of light.

Chapter 8 - Heat and Thermodynamics

Questions 24-40 | 17 questions | 3 marks each

24. What is thermal equilibrium? Explain what happens when two bodies at different temperatures are brought into thermal contact.

- Two bodies are in thermal equilibrium when they have the same temperature and no net heat flows between them.
- If their temperatures differ, heat flows spontaneously from the hotter body to the colder body.
- This transfer continues until their temperatures become equal.

25. Explain the relationship between temperature and the internal energy of a substance.

- Internal energy is the total microscopic kinetic and potential energy of the particles of a substance.
- A temperature rise generally increases the particles' average random kinetic energy.
- Therefore, heating a substance without a change of state generally increases both its temperature and internal energy.

26. State three basic assumptions of the kinetic theory of an ideal gas.

- A gas contains a very large number of molecules whose individual volumes are negligible compared with the container's volume.
- Molecules move continuously and randomly, exerting negligible intermolecular forces except during collisions.
- Collisions between molecules and with the container walls are perfectly elastic.

27. Under what conditions does a real gas behave approximately like an ideal gas? Explain why.

- A real gas behaves most nearly ideally at low pressure and sufficiently high temperature, away from condensation.
- At low pressure, molecules are widely separated, making their own volume relatively unimportant.
- At high temperature, their kinetic energy is large compared with the effects of intermolecular attraction.

28. Explain why the pressure of a gas increases with temperature at constant volume.

- Increasing temperature raises the average kinetic energy and speed of gas molecules.
- The molecules strike the container walls more frequently and transfer more momentum in each collision on average.
- Therefore, the force per unit area increases, so gas pressure rises.

29. Explain why a gas expands when heated at constant pressure.

- Heating increases the average speed and kinetic energy of the gas molecules.
- The gas expands, reducing the frequency of molecular collisions with the walls per unit area.
- This compensates for the increased molecular motion, allowing pressure to remain constant while volume increases.

30. Explain Boyle's law using the kinetic theory of gases.

- At constant temperature, the average kinetic energy of the gas molecules remains unchanged.
- Decreasing the volume makes the molecules collide with the walls more frequently.
- Pressure therefore increases as volume decreases, giving $PV = \text{constant}$ for a fixed amount of gas.

31. Distinguish between work done by a gas and work done on a gas.

- During expansion, the gas pushes its surroundings and does work on them.
- During compression, the surroundings do work on the gas.
- Using the convention that work done **by** the gas is positive, $W = \int P\,dV$: expansion gives positive W, while compression gives negative W.

32. State the first law of thermodynamics and explain how it expresses conservation of energy.

- The first law states that heat supplied to a system equals its increase in internal energy plus work done by it.
- Mathematically, $Q = \Delta U + W$, where W is work done by the system.
- Supplied energy is accounted for as stored internal energy or transferred work, so energy is conserved.

33. Distinguish between isothermal and adiabatic processes.

- In an isothermal process, temperature remains constant; in an adiabatic process, no heat is exchanged with the surroundings.
- For an ideal gas undergoing an isothermal process, $\Delta U = 0$, so $Q = W$.
- During adiabatic expansion, the gas does work using its internal energy, so its temperature decreases.

34. Compare isochoric and isobaric heating of a gas.

- Isochoric heating occurs at constant volume; isobaric heating occurs at constant pressure.
- In isochoric heating, no expansion work is done, so $Q = \Delta U$.
- In isobaric heating, the gas expands and does work, so $Q = \Delta U + P\Delta V$.

35. State the working principle of a heat engine.

- A heat engine absorbs heat Q_H from a high-temperature reservoir.
- It converts part of this heat into useful work W and rejects the remaining heat Q_C to a colder reservoir.
- The working substance returns to its initial state after each cycle, with $W = Q_H - Q_C$.

36. Distinguish between reversible and irreversible processes.

- A reversible process can be reversed so that both the system and surroundings return to their original states.
- An irreversible process cannot be reversed without leaving an overall change in the system or surroundings.
- Ideal, extremely slow frictionless processes can approach reversibility; friction and heat flow through a finite temperature difference cause irreversibility.

37. State and explain the second law of thermodynamics.

- A cyclic heat engine cannot convert all the heat taken from a single reservoir into work.
- Heat cannot flow spontaneously from a colder body to a hotter body without external work.
- Thus, a heat engine must reject some heat, while a refrigerator requires a work input.

38. State the principle of a Carnot engine and explain its efficiency limit.

- A Carnot engine is an ideal reversible engine operating through two isothermal and two adiabatic processes.
- No engine operating between the same two reservoir temperatures can be more efficient.
- Its maximum efficiency is $\eta = 1 - T_C/T_H$, with both temperatures expressed in kelvin.

39. Explain why a refrigerator is described as a heat engine operating in reverse.

- A refrigerator removes heat Q_C from a cold region and transfers heat to a warmer region.
- External work W, supplied to the compressor, makes this transfer possible.
- It rejects $Q_H = Q_C + W$ to the surroundings, reversing the usual direction of heat transfer in a heat engine.

40. Explain entropy and the degradation of energy in natural processes.

- Entropy is a thermodynamic quantity associated with the spreading of energy and the disorder of a system.
- In a spontaneous irreversible process, the total entropy of an isolated system increases.
- Energy remains conserved, but more of it becomes unavailable for useful work; this is called energy degradation.

Chapter 12 - Magnetism / Electromagnetism

Questions 41-55 | 15 questions | 3 marks each

41. Define magnetic flux density and state its SI unit.

- Magnetic flux density is the force per unit current per unit length on a straight conductor placed perpendicular to a magnetic field.
- It is given by $B = F/(IL)$ for this perpendicular arrangement.
- Its SI unit is tesla: $1\,\mathrm{T} = 1\,\mathrm{N\,A^{-1}\,m^{-1}}$.

42. Describe the force on a current-carrying conductor placed in a magnetic field.

- A conductor carrying current I through length L in a magnetic field experiences force $F = BIL\sin\theta$.
- The force is maximum when the conductor is perpendicular to the field and zero when it is parallel.
- Its direction is perpendicular to both current and field and is found using Fleming's left-hand rule.

43. Describe the motion of a charged particle entering a uniform magnetic field perpendicular to the field.

- The particle experiences a magnetic force of magnitude $F = |q|vB$.
- The force remains perpendicular to its velocity and acts as a centripetal force.
- The particle therefore follows a circular path with radius $r = mv/(|q|B)$, at non-relativistic speeds.

44. Why does a magnetic field do no work on a moving charged particle?

- Magnetic force is perpendicular to the particle's instantaneous velocity.
- Therefore, it has no component along the instantaneous displacement.
- The work done is zero, so the field changes the direction of motion without changing speed or kinetic energy.

45. Explain how crossed electric and magnetic fields act as a velocity selector.

- Electric and magnetic fields are arranged perpendicular to each other and to the incoming particle beam.
- Their forces are arranged to oppose each other, so an undeflected particle satisfies $qE = qvB$.
- Only particles with speed $v = E/B$ pass straight through; particles with other speeds are deflected.

46. Describe the magnetic field around a long straight current-carrying wire.

- Magnetic field lines form concentric circles centred on the wire.
- Their direction follows the right-hand grip rule: thumb along current, curled fingers along the field.
- The field becomes stronger with increasing current and weaker with increasing distance from the wire.

47. Describe the magnetic field produced by a current-carrying circular coil.

- The magnetic fields produced by different parts of the coil reinforce one another near its centre.
- At the centre, the resultant field is perpendicular to the plane of the coil.
- Its direction follows the right-hand grip rule, and its strength increases with current and number of turns.

48. Describe the magnetic field pattern of a long current-carrying solenoid.

- The field resembles that of a bar magnet, with a north pole at one end and a south pole at the other.
- Inside a long solenoid, field lines are close together, approximately parallel and uniformly spaced.
- Outside, the field is weaker and the lines curve around to form closed loops.

49. Why does inserting a soft-iron core increase the magnetic field of a solenoid?

- Soft iron contains magnetic domains that become aligned in the solenoid’s magnetic field.
- The magnetized core produces an additional field in the same direction.
- The combined magnetic field becomes much stronger than that of the solenoid alone.

50. Explain the forces between two parallel current-carrying conductors.

- Each conductor produces a magnetic field at the position of the other.
- Each current therefore experiences a magnetic force due to the other conductor’s field.
- Currents in the same direction attract, while currents in opposite directions repel.

51. Distinguish between magnetic flux and magnetic flux linkage.

- Magnetic flux measures the magnetic field passing through a surface: $\Phi = BA\cos\theta$.
- Flux linkage is the total linked flux for a coil; if every turn links the same flux, it equals $N\Phi$.
- Magnetic flux is measured in weber, while flux linkage is commonly expressed in weber-turns.

52. Describe a simple experiment demonstrating electromagnetic induction.

- Connect a coil to a sensitive galvanometer and move a bar magnet towards the coil.
- The galvanometer deflects while the magnetic flux through the coil changes, showing an induced current.
- Withdrawing the magnet reverses the deflection; keeping both magnet and coil stationary produces no sustained deflection.

53. State Lenz’s law and explain its connection with conservation of energy.

- Lenz’s law states that an induced current flows in a direction that opposes the change producing it.
- An external agent must therefore do work against the opposing magnetic effect.
- This mechanical work supplies the electrical energy produced, in agreement with conservation of energy.

54. State three ways of increasing the induced e.m.f. in a coil.

- Increase the number of turns, provided each turn links the changing magnetic flux.
- Move the magnet or coil faster so that flux changes more rapidly.
- Use a stronger magnetic field for the same relative motion, increasing the rate of change of flux linkage.

55. Explain how a seismometer uses electromagnetic induction to detect earthquakes.

- Ground vibrations move the instrument’s frame, producing relative motion between its suspended coil and magnet.
- This changes the magnetic flux linkage and induces an e.m.f. according to Faraday’s law.
- The electrical signal is recorded against time to provide a record of the ground vibrations.

Chapter 11 - Electricity

Questions 56-69 | 14 questions | 3 marks each

56. What is drift velocity? Explain its relationship with electric current.

- Drift velocity is the average velocity acquired by charge carriers under an applied electric field.
- The current magnitude is $I = nA|q|v_d$, where n is carrier number density and A is cross-sectional area.
- In metals, electrons drift opposite to the direction of conventional current.

57. Explain potential difference and its relationship with electrical power.

- Potential difference is the energy transferred per unit charge between two points: $V = W/Q$.
- Electrical power is the energy transferred per unit time, giving $P = VI$.
- For an ohmic resistor, this can also be written as $P = I^2R = V^2/R$.

58. Explain the factors affecting the resistance of a uniform wire.

- Resistance increases with wire length because charge carriers experience opposition over a longer path.
- Resistance decreases with increasing cross-sectional area because more conducting paths are available.
- $R = \rho L/A$, where resistivity ρ depends on the material and temperature.

59. How does an LDR act as a light-sensitive potential divider?

- An LDR is a resistor whose resistance decreases as incident light intensity increases.
- Connecting it in series with a fixed resistor divides the supply voltage between them.
- Changing light intensity changes the resistance ratio and therefore the output voltage, allowing light-sensitive control.

60. Distinguish between e.m.f. and potential difference.

- E.m.f. is the energy supplied by a source per unit charge in driving charge around a complete circuit.
- Potential difference across a component is the energy transferred per unit charge between its terminals.
- Both are measured in volts, but a cell’s terminal potential difference during discharge can be less than its e.m.f.

61. Explain how internal resistance affects the terminal potential difference of a cell.

- A real cell has internal resistance r, which causes an internal voltage drop Ir when current flows.
- During discharge, terminal potential difference is $V = \mathcal{E} - Ir$.
- A larger current causes a greater internal voltage drop, reducing the voltage available to the external circuit.

62. State Kirchhoff's first law and explain its physical basis.

- The total current entering a junction equals the total current leaving it.
- Equivalently, the algebraic sum of currents at a junction is zero.
- This follows from conservation of charge because charge does not continuously accumulate at an ideal circuit junction.

63. State Kirchhoff's second law and explain its physical basis.

- The algebraic sum of all potential changes around a closed circuit loop is zero.
- Thus, the total voltage rise equals the total voltage drop around the loop.
- This follows from conservation of energy: charge returning to its starting point has no net change in electric potential energy.

64. Explain why resistances add when resistors are connected in series.

- The same current passes through every resistor because there is only one conducting path.
- The total potential difference equals the sum of the individual potential differences.
- Using $V_i = IR_i$, the equivalent resistance is $R_{\text{eq}} = R_1 + R_2 + \cdots$.

65. Explain why the equivalent resistance of parallel resistors is less than the smallest individual resistance.

- Every parallel branch has the same potential difference across it.
- Adding a branch provides an additional path for current, increasing the total current for the same voltage.
- Therefore, equivalent resistance decreases, with $1/R_{\text{eq}} = 1/R_1 + 1/R_2 + \cdots$.

66. State the principle of a potentiometer used to compare potential differences.

- Along a uniform wire carrying a constant current, potential drop is proportional to wire length.
- Each potential difference is balanced against the drop along a measured length of the wire.
- The ratio of the potential differences equals the ratio of their balancing lengths: $V_1/V_2 = l_1/l_2$.

67. Explain the role of a galvanometer in a null method of measurement.

- The galvanometer detects a small current in the comparison branch.
- The circuit is adjusted until its deflection becomes zero, indicating a balance condition.
- At balance, no current flows through that branch, allowing the unknown quantity to be found from the known circuit relationships.

68. Explain how an NTC thermistor is used in a temperature-sensitive potential divider.

- The resistance of an NTC thermistor decreases as its temperature rises.
- When connected in series with a fixed resistor, its changing resistance changes the division of the supply voltage.
- The resulting output voltage can operate a temperature detector, alarm or control circuit.

69. How can carbon fibres help inspectors check the condition of a concrete bridge?

- Conducting carbon fibres form electrical paths within the concrete.
- Cracks or deformation can disturb these paths and increase their electrical resistance.
- Monitoring resistance changes helps inspectors detect developing damage and assess the bridge’s condition.

Chapter 6 - Fluid Mechanics

Questions 70-81 | 12 questions | 3 marks each

70. State Archimedes' principle and explain the condition for flotation.

- A body immersed partly or completely in a fluid experiences an upward force equal to the weight of fluid displaced.
- A freely floating body is in equilibrium when its upthrust equals its weight.
- Therefore, a floating body displaces a weight of fluid equal to its own weight.

71. Why can a steel ship float even though solid steel is denser than water?

- A ship has a hollow hull enclosing a large volume of air.
- Its overall average density is lower than that of solid steel and can be less than the density of water.
- It sinks into the water until the weight of displaced water equals the ship's total weight.

72. Describe the main characteristics of ideal-fluid flow.

- The flow is steady: fluid velocity at a particular point does not change with time.
- The fluid is incompressible, so its density remains constant.
- It is non-viscous, so internal friction between neighbouring fluid layers is absent.

73. Explain why the equation of continuity expresses conservation of mass.

- In steady flow, mass cannot continuously accumulate within a section of a pipe.
- Therefore, the mass entering per second equals the mass leaving per second: $\rho_1 A_1 v_1 = \rho_2 A_2 v_2$.
- For an incompressible fluid, density is constant, giving $A_1 v_1 = A_2 v_2$.

74. Why does water emerge faster when the end of a rubber pipe is squeezed?

- Squeezing reduces the cross-sectional area of the outlet.
- For a given steady volume flow rate of incompressible water, $Av = \text{constant}$.
- Therefore, the smaller outlet area requires a greater flow speed.

75. State Bernoulli's equation and explain the energy represented by its terms.

- For steady, incompressible, non-viscous flow along a streamline, $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$.
- The three terms represent pressure energy, kinetic energy and gravitational potential energy per unit volume.
- The equation expresses conservation of mechanical energy in ideal-fluid flow.

76. Explain the working of a perfume atomizer using Bernoulli’s principle.

- Fast-moving air passes over the upper end of a tube dipping into the perfume.
- The air pressure at the tube’s upper end becomes lower than the pressure acting on the liquid surface.
- This pressure difference raises the perfume through the tube, and the air breaks it into fine droplets.

77. Explain the principle of a Venturi meter.

- A Venturi meter contains a narrow section within a wider pipe.
- Fluid speeds up in the narrow section according to continuity, and its pressure decreases according to Bernoulli’s equation.
- Measuring the pressure difference allows the flow speed or flow rate to be determined.

78. Explain the Magnus effect on a spinning ball moving through air.

- The spinning ball influences the surrounding airflow, producing different air speeds on opposite sides.
- The resulting pressure difference produces a sideways or lifting force on the ball.
- This force bends the ball’s path; its direction depends on the spin direction and motion through the air.

79. What is viscosity? Why are real fluids viscous?

- Viscosity is the internal resistance of a fluid to relative motion between neighbouring layers.
- Molecular interactions and the transfer of momentum between layers oppose their relative movement.
- Therefore, real fluids experience internal friction and dissipate mechanical energy during flow.

80. Explain why a body moving through a fluid experiences a retarding force.

- The moving body interacts with the fluid and causes the surrounding fluid to move or deform.
- Viscous effects and pressure differences produce a drag force opposing the body’s motion relative to the fluid.
- Drag depends on factors such as speed, fluid viscosity, and the body’s size and shape.

81. What is superfluidity? State two consequences of this state.

- Superfluidity is a state in which a fluid component flows without viscosity.
- A superfluid film can creep along container surfaces and over their walls.
- Once established, circulation can persist for extremely long periods because viscous resistance is absent.

Chapter 4 - Rotational and Circular Motion

Questions 82-94 | 13 questions | 3 marks each

82. Define angular displacement, angular velocity and angular acceleration.

- Angular displacement θ is the angle through which a body rotates about an axis; its SI unit is radian.
- Angular velocity is the rate of change of angular displacement: $\omega = d\theta/dt$.
- Angular acceleration is the rate of change of angular velocity: $\alpha = d\omega/dt$.

83. Why is a body in uniform circular motion accelerated even though its speed is constant?

- Velocity depends on both magnitude and direction.
- During circular motion, its direction changes continuously even when its magnitude remains constant.
- Therefore, the body has centripetal acceleration directed towards the centre: $a_c = v^2/r$.

84. Explain how friction and the normal force can provide centripetal force to a turning vehicle.

- On a level road, static friction between the tyres and road supplies the horizontal centripetal force.
- On a banked road, the normal force has a horizontal component directed towards the centre of the turn.
- This component reduces the reliance on friction and helps the vehicle turn safely.

85. Explain the relationships between linear and angular quantities for a rotating body.

- A point at radius r travels an arc length $s = r\theta$.
- Its tangential speed is $v = r\omega$, so points farther from the axis move faster for the same angular velocity.
- Its tangential acceleration is $a_t = r\alpha$, while its centripetal acceleration is $a_c = r\omega^2$.

86. What is moment of inertia? Explain the factors on which it depends.

- Moment of inertia measures a body’s resistance to a change in rotational motion about a specified axis.
- For particles, $I = \sum mr^2$, so it depends on both mass and distance from the axis.
- Moving mass farther from the axis increases the moment of inertia, even if the total mass is unchanged.

87. State the relationship between torque and angular acceleration and explain its meaning.

- For a rigid body rotating about a fixed axis with constant moment of inertia, $\tau_{\text{net}} = I\alpha$.
- For a given moment of inertia, a larger net torque produces a larger angular acceleration.
- For a given torque, a body with a larger moment of inertia has a smaller angular acceleration.

88. State the law of conservation of angular momentum and its condition.

- Angular momentum remains constant when the net external torque on a system is zero.
- For rotation about a fixed axis, angular momentum is $L = I\omega$.
- Therefore, $I_1\omega_1 = I_2\omega_2$, allowing angular velocity to change when moment of inertia changes.

89. Why does an ice skater spin faster when the arms are pulled inward?

- Pulling the arms inward brings more body mass closer to the axis of rotation.
- This reduces the skater's moment of inertia.
- With negligible external torque, angular momentum is conserved, so angular velocity increases.

90. Explain how a flywheel stores energy and helps machinery run smoothly.

- A flywheel is a rotating wheel or disc with a relatively large moment of inertia.
- It stores rotational kinetic energy when energy is supplied to speed it up.
- It releases stored energy when the driving input falls, reducing sudden changes in the machine's speed.

91. How does a gyroscope help maintain a reference direction?

- A rapidly spinning gyroscope possesses angular momentum along its rotation axis.
- Changing the direction of this angular momentum requires an external torque.
- With small external torques, its axis remains approximately fixed, making it useful in navigation and stabilization.

92. Why do astronauts appear weightless in an orbiting satellite?

- Gravity acts on both the satellite and the astronauts, keeping them in continuous free fall around Earth.
- They fall together with approximately the same acceleration.
- The supporting normal force is nearly zero, so their apparent weight is nearly zero even though gravity is present.

93. How can artificial gravity be produced in a spacecraft?

- A spacecraft or part of it can be rotated about an axis.
- Its outer wall provides the inward centripetal force needed to keep occupants moving in a circle.
- Occupants feel this contact force as apparent weight, with the required acceleration given by $a = \omega^2 r$.

94. Explain the principle of separation in a centrifuge.

- A centrifuge rapidly rotates a mixture, requiring its components to undergo centripetal acceleration.
- Components sediment differently according to factors such as density and particle size; denser suspended components generally collect farther from the axis.
- This allows materials to be separated, for example blood cells from plasma.

Chapter 14 - Particle Physics

Questions 95-106 | 12 questions | 3 marks each

95. Describe the nature, charge and mass of alpha, beta and gamma radiations.

- An alpha particle is a helium nucleus containing two protons and two neutrons; its charge is $+2e$ and mass approximately $4u$.
- Beta particles are electrons (β^-) or positrons (β^+), with charges $-e$ and $+e$, respectively, and equal small masses.
- Gamma radiation consists of high-energy photons with no electric charge and zero rest mass.

96. Explain conservation of nucleon number and charge in nuclear processes.

- The total nucleon number before a nuclear process equals the total nucleon number afterwards.
- Total electric charge is also conserved, including the charge carried by emitted particles.
- Therefore, both mass numbers and charge numbers must balance when a nuclear equation is written.

97. What is an antiparticle? Explain electron-positron annihilation.

- An antiparticle has the same mass as its corresponding particle but opposite electric charge when the particle is charged.
- A positron is the antiparticle of an electron; when they meet, they can annihilate.
- In the usual two-photon process, their energy becomes gamma-ray photons while total energy, momentum and charge remain conserved.

98. Describe the particles emitted in beta-minus and beta-plus decay.

- In beta-minus decay, a neutron becomes a proton, an electron and an electron antineutrino: $n \rightarrow p + e^- + \bar{\nu}_e$.
- In beta-plus decay inside a suitable nucleus, a proton becomes a neutron, a positron and an electron neutrino: $p \rightarrow n + e^+ + \nu_e$.
- The emitted neutrino or antineutrino carries away part of the released energy and momentum.

99. Why do alpha particles have discrete energies, whereas beta particles have a continuous energy spectrum?

- A particular alpha transition occurs between definite nuclear energy states, giving a definite released energy.
- In beta decay, the released energy is shared among the beta particle, the neutrino or antineutrino, and the recoiling nucleus.
- The beta particle can receive different shares, so its energy varies continuously up to a maximum.

100. What are quarks and antiquarks? Name the six quark flavours.

- Quarks are fundamental particles that combine to form hadrons such as protons and neutrons.
- The six flavours are up, down, charm, strange, top and bottom.
- Every quark has a corresponding antiquark with the same mass and opposite electric charge.

101. Describe the quark composition and charge of a proton and a neutron.

- A proton contains two up quarks and one down quark: uud.
- A neutron contains one up quark and two down quarks: udd.
- Since an up quark has charge $+2e/3$ and a down quark has charge $-e/3$, the proton has charge $+e$ and the neutron has zero net charge.

102. Distinguish between baryons and mesons.

- Both baryons and mesons are hadrons and participate in strong interactions.
- A baryon consists of three quarks; examples include the proton and neutron.
- A meson consists of a quark-antiquark pair; examples include pions and kaons.

103. Explain the changes in quark composition during beta decay.

- In beta-minus decay, a down quark changes into an up quark, converting a neutron (udd) into a proton (uud).
- In beta-plus decay, an up quark changes into a down quark, converting a proton into a neutron.
- These changes occur through the weak interaction, with an electron-antineutrino pair or positron-neutrino pair emitted, respectively.

104. What are leptons? Give examples and distinguish them from hadrons.

- Leptons are fundamental particles that are not composed of quarks.
- Examples include electrons, muons, tau particles and their associated neutrinos.
- Leptons do not experience the strong interaction, whereas hadrons do.

105. Identify the exchange particles for electromagnetic, strong and weak interactions.

- Photons mediate electromagnetic interactions between electrically charged particles.
- Gluons mediate the strong interaction between quarks.
- The W^+, W^- and Z^0 bosons mediate weak interactions.

106. Describe the Higgs boson and its connection with particle mass.

- The Higgs boson is a fundamental, electrically neutral particle with zero spin.
- It is an excitation of the Higgs field, which is present throughout space.
- Interaction with the Higgs field gives mass to elementary particles such as electrons and quarks; different interaction strengths produce different masses.

Chapter 13 - Relativity

Questions 107-115 | 9 questions | 3 marks each

107. Distinguish between inertial and non-inertial frames of reference.

- An inertial frame is one in which a force-free body remains at rest or moves with constant velocity.
- A frame accelerating or rotating relative to an inertial frame is non-inertial.
- Newton’s laws apply directly in an inertial frame; additional apparent forces are needed when applying them in a non-inertial frame.

108. State the two postulates of special relativity and give their significance.

- The laws of physics are the same in all inertial frames of reference.
- The speed of light in vacuum is the same for all inertial observers, independent of the motion of source or observer.
- These postulates require measurements of time and length to depend on relative motion.

109. Why is the constancy of the speed of light different from ordinary addition of velocities?

- In ordinary low-speed mechanics, relative velocities are commonly found by simple addition or subtraction.
- Light in vacuum is measured to travel at c by every inertial observer, regardless of relative motion.
- Therefore, ordinary velocity addition cannot be applied to light; relativistic relationships must be used.

110. Explain why space and time become relative if the speed of light is constant.

- Different inertial observers must all measure the same vacuum speed of light.
- Their measurements of distance and elapsed time therefore cannot both remain identical under relative motion.
- Space and time measurements change together, leading to effects such as length contraction and time dilation.

111. What is meant by relativity of simultaneity?

- Two events are simultaneous in a frame if they occur at the same time according to clocks synchronized in that frame.
- Events simultaneous in one inertial frame may occur at different times in another moving frame.
- Therefore, simultaneity of spatially separated events is not absolute; it depends on the observer’s frame.

112. Explain time dilation and define proper time.

- Proper time Δt_0 is the interval measured by a clock present at both events.
- An observer relative to whom that clock moves measures a longer interval: $\Delta t = \gamma \Delta t_0$.
- Thus, a moving clock is observed to run more slowly, where $\gamma = 1/\sqrt{1 - v^2/c^2}$.

113. Explain length contraction and define proper length.

- Proper length L_0 is the length of an object measured in its own rest frame.
- An observer relative to whom the object moves measures a shorter length along the direction of motion: $L = L_0/\gamma$.
- Dimensions perpendicular to the motion are unchanged by this contraction.

114. State mass-energy equivalence and explain its significance.

- A body with rest mass m_0 has rest energy $E_0 = m_0 c^2$.
- A change in a system's rest mass corresponds to an energy change $\Delta E = \Delta m\, c^2$.
- This explains why a small decrease in rest mass can release a large amount of energy in nuclear processes.

115. Explain relativistic mass variation and why a massive particle cannot reach the speed of light.

- In the textbook's relativistic-mass convention, $m = \gamma m_0$, while the rest mass m_0 remains unchanged.
- As speed approaches c, γ and the particle's total energy increase without limit.
- Reaching c would require infinite energy, so a particle with non-zero rest mass cannot be accelerated to that speed.

Chapter 5 - Work and Kinetic Energy

Questions 116-123 | 8 questions | 3 marks each

116. How is work determined from a force-displacement graph?

- Plot the component of force along the displacement against displacement.
- The signed area between the graph and the displacement axis gives the work done.
- For constant force, the area is rectangular, giving $W = F_{\parallel} s$.

117. Explain how a force-displacement graph gives work done by a variable force.

- Divide the displacement into small intervals over which the force is approximately constant.
- The work in each interval is approximately the force component multiplied by that interval's displacement.
- Adding these contributions gives the area under the graph, representing total work.

118. State the main properties of a conservative force and give examples.

- Work done between two points is independent of the path taken.
- The total work done around a closed path is zero.
- Gravitational and electrostatic forces are examples; potential energy can be associated with them.

119. What is a non-conservative force? Explain with an example.

- A non-conservative force does work that depends on the path followed.
- Its total work around a closed path is generally not zero.
- Friction is an example because it converts mechanical energy into internal energy over the distance travelled.

120. State the work-energy theorem and explain its meaning.

- The work-energy theorem states that net work done on a body equals its change in kinetic energy.
- Mathematically, $W_{\text{net}} = K_f - K_i$.
- Positive net work increases kinetic energy, while negative net work decreases it.

121. Explain the work-energy theorem for a body moving through a resistive medium.

- An applied force transfers energy to the body, while resistance opposes its motion.
- Part of the applied work increases kinetic energy, and part is transferred against resistance.
- Thus, $W_{\text{applied}} = \Delta K + E_{\text{dissipated}}$, where dissipated energy usually becomes internal energy.

122. Why does friction reduce mechanical energy without violating conservation of energy?

- Friction generally acts opposite to relative motion and does negative work on the moving body.
- Mechanical energy is transferred into internal energy of the contacting surfaces and sometimes sound.
- Total energy remains conserved, although less energy remains as kinetic or potential energy.

123. Define kinetic energy and explain why it is a scalar quantity.

- Kinetic energy is the energy possessed by a body because of its motion.
- At non-relativistic speeds, $K = \frac{1}{2}mv^2$, so it depends on mass and the square of speed.
- Mass and squared speed have no direction, so kinetic energy is scalar and non-negative.

Chapter 3 - Translatory Motion

Questions 124-131 | 8 questions | 3 marks each

124. Explain the independence of horizontal and vertical motions of a projectile.

- Neglecting air resistance, gravity acts vertically, producing no horizontal acceleration.
- The horizontal velocity therefore remains constant, while the vertical velocity changes under gravity.
- The two components can be analysed separately, using the same elapsed time.

125. Describe the velocity and acceleration of a projectile at its highest point.

- Its vertical velocity is momentarily zero.
- Its horizontal velocity remains unchanged if air resistance is neglected.
- Its acceleration is still g vertically downward; it is not zero at the highest point.

126. State the condition for maximum projectile range and explain complementary launch angles.

- For launch and landing at the same height without air resistance, $R = u^2 \sin 2\theta / g$.
- For a fixed launch speed, range is maximum when $\theta = 45^\circ$.
- Complementary launch angles produce equal ranges because they give equal values of $\sin 2\theta$.

127. Explain how air resistance affects projectile motion.

- Air resistance acts opposite to the projectile’s instantaneous velocity.
- It reduces horizontal speed and makes the trajectory asymmetric rather than an ideal parabola.
- For the same launch conditions, it generally reduces maximum height and horizontal range.

128. Why do bodies of different masses fall with the same acceleration in the absence of air resistance?

- The gravitational force on a body is proportional to its mass: $F = mg$.
- Its resistance to acceleration is also proportional to mass, since $F = ma$.
- Combining the relations gives $a = g$, independent of the body’s mass.

129. Distinguish between elastic and inelastic collisions.

- Total momentum is conserved in both types if the system experiences no net external impulse.
- Total kinetic energy is conserved in an elastic collision.
- In an inelastic collision, some kinetic energy changes into other forms; in a perfectly inelastic collision, the bodies stick together.

130. Why can momentum remain conserved in a collision while kinetic energy changes?

- Momentum conservation follows from the absence of a net external impulse on the system.
- Internal interactions can transform kinetic energy into deformation, internal energy or sound.
- Therefore, total momentum may remain unchanged even though total kinetic energy decreases.

131. What happens in a head-on elastic collision between two equal masses when one is initially at rest?

- Both total momentum and total kinetic energy are conserved.
- For equal masses in a one-dimensional elastic collision, the bodies exchange velocities.
- The initially moving body stops, and the initially stationary body moves with the first body’s original velocity.

Chapter 2 - Vectors

Questions 132-138 | 7 questions | 3 marks each

132. What are rectangular components of a vector? Explain how they are obtained.

- Rectangular components are the mutually perpendicular vectors whose sum equals the original vector.
- For a vector of magnitude A at angle θ to the positive x-axis, $A_x = A\cos\theta$.
- Its y-component is $A_y = A\sin\theta$, so $\vec{A} = A_x\hat{i} + A_y\hat{j}$.

133. Explain how the magnitude and direction of a vector are obtained from its components.

- Perpendicular components form the two sides of a right-angled triangle.
- The magnitude follows from Pythagoras' theorem: $A = \sqrt{A_x^2 + A_y^2}$.
- The direction satisfies $\tan\theta = A_y/A_x$, with the correct quadrant determined from the component signs.

134. Define scalar product and give two physical examples.

- The scalar product is $\vec{A}\cdot\vec{B} = AB\cos\theta$, where θ is the angle between the vectors.
- It gives a scalar result, such as work: $W = \vec{F}\cdot\vec{s}$.
- Another example is instantaneous power: $P = \vec{F}\cdot\vec{v}$.

135. State three important properties of the scalar product.

- It is commutative: $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}$.
- The dot product of perpendicular vectors is zero.
- The dot product of a vector with itself equals the square of its magnitude: $\vec{A}\cdot\vec{A} = A^2$.

136. Define vector product and explain its direction.

- The vector product is $\vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$.
- Its direction is perpendicular to the plane containing the two vectors.
- The right-hand rule determines its direction; reversing the order reverses the result.

137. State three important properties of the vector product.

- It is anti-commutative: $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$.
- The cross product of parallel or antiparallel vectors is zero.
- For fixed magnitudes, its magnitude is maximum when the vectors are perpendicular.

138. Distinguish between scalar and vector products using physical examples.

- A scalar product produces a scalar; a vector product produces a vector.
- Their magnitudes involve $\cos\theta$ and $\sin\theta$, respectively.
- Work $W = \vec{F} \cdot \vec{s}$ is a scalar-product example, while torque $\vec{\tau} = \vec{r} \times \vec{F}$ is a vector-product example.

Chapter 10 - Electrostatics

Questions 139-145 | 7 questions | 3 marks each

139. Define electric field strength and explain its direction.

- Electric field strength is the electric force per unit positive test charge: $\vec{E} = \vec{F}/q$.
- Its direction is the direction of force on a positive test charge.
- Its SI unit is $\mathrm{N\,C^{-1}}$, equivalent to $\mathrm{V\,m^{-1}}$.

140. State three properties of electric field lines.

- Electric field lines point away from positive charges and towards negative charges.
- The tangent to a field line gives the electric field direction at that point.
- Field lines never intersect, and closer spacing represents a stronger electric field.

141. Describe the motion of a charged particle entering a uniform electric field perpendicular to the field.

- The particle experiences a constant force $\vec{F} = q\vec{E}$ and therefore constant acceleration along or opposite to the field.
- Its velocity component perpendicular to the field remains constant, neglecting other forces.
- The combined motion gives a parabolic path; positive and negative particles deflect in opposite directions.

142. How is the electric field outside a charged spherical conductor represented?

- Outside an isolated spherical conductor, the field is the same as if its total charge were concentrated at its centre.
- The field is radial and has magnitude $E = k|Q|/r^2$, where r is measured from the centre.
- It points outward for positive charge and inward for negative charge.

143. Explain how a Faraday cage provides electrostatic shielding.

- A Faraday cage is a conducting enclosure in which free charges redistribute under an external electric field.
- Their induced electric field opposes the external field inside the enclosure.
- In electrostatic equilibrium, an enclosed cavity containing no charge is shielded from the external electric field.

144. State Coulomb’s law and describe the direction of the force.

- The force between two stationary point charges is proportional to the product of their charge magnitudes.
- It is inversely proportional to the square of their separation: $F = k|q_1 q_2|/r^2$.
- The force acts along the line joining the charges: like charges repel and unlike charges attract.

145. Explain the electric field between oppositely charged parallel plates and its relation to potential difference.

- Away from the edges, large closely spaced parallel plates produce an approximately uniform electric field.
- Field lines are straight, parallel and directed from the positive plate to the negative plate.
- The field magnitude is $E = |\Delta V|/d$, where d is the plate separation.

Chapter 1 - Physical Quantities and Measurements

Questions 146-150 | 5 questions | 3 marks each

146. What are derived units? Express the units of force and pressure in SI base units.

- A derived unit is formed by combining SI base units according to the defining physical relationship.
- From $F = ma$, the unit of force is newton: $\mathrm{N} = \mathrm{kg\,m\,s^{-2}}$.
- From $P = F/A$, the unit of pressure is pascal: $\mathrm{Pa} = \mathrm{kg\,m^{-1}\,s^{-2}}$.

147. State the principle of dimensional homogeneity and explain a limitation of dimensional analysis.

- Every term added or equated in a physically valid equation must have the same dimensions.
- Dimensional analysis can therefore identify equations whose terms have inconsistent dimensions.
- Dimensional consistency alone cannot prove an equation correct or determine dimensionless numerical constants.

148. Distinguish between accuracy and precision in measurement.

- Accuracy describes how close a measured result is to the accepted or true value.
- Precision describes how closely repeated measurements agree with one another.
- Measurements can be precise but inaccurate if a systematic error shifts all readings away from the true value.

149. Why does every physical measurement contain some uncertainty?

- Every measuring instrument has limited resolution, so it cannot distinguish arbitrarily small changes.
- Instrumental imperfections, environmental variations and the observer’s technique can affect readings.
- Therefore, a result should include an uncertainty indicating the reasonable range of the measured value.

150. State the rules for combining uncertainties in derived quantities.

- For addition or subtraction, add the absolute uncertainties to estimate the maximum uncertainty.
- For multiplication or division, add fractional or percentage uncertainties.
- For a power $y = x^n$, multiply the fractional uncertainty in x by $|n|$.

Chapter 7 - Physics of Solids

Questions 151-156 | 6 questions | 3 marks each

151. Distinguish between crystalline and amorphous solids.

- Crystalline solids have a regular, repeating arrangement with long-range order; amorphous solids lack long-range order.
- Crystalline solids generally have a definite melting temperature, while amorphous solids soften over a range of temperatures.
- Common salt is crystalline, whereas ordinary glass is amorphous.

152. What are polymeric solids? State their important characteristics.

- Polymeric solids consist of large molecules formed from many repeating smaller units called monomers.
- Their molecules usually form long chains, with properties depending on chain arrangement and bonding.
- Examples include polythene, PVC and rubber; many polymers have low density and are poor electrical conductors.

153. Define tensile stress, tensile strain and Young’s modulus.

- Tensile stress is the stretching force per unit cross-sectional area: $\sigma = F/A$.
- Tensile strain is the extension divided by original length: $\varepsilon = \Delta L/L$.
- Young’s modulus is the ratio of tensile stress to tensile strain in the proportional elastic region: $Y = \sigma/\varepsilon$.

154. Outline an experiment to determine Young’s modulus of a metal wire.

- Measure the wire’s original length L and diameter, using the diameter to find cross-sectional area A.
- Add known loads gradually and measure the corresponding extensions ΔL, remaining within the proportional elastic region.
- Determine Young’s modulus from $Y = FL/(A\Delta L)$, or from the gradient of the load-extension graph.

155. Distinguish between elastic and plastic deformation and define the elastic limit.

- Elastic deformation disappears when the deforming force is removed, restoring the original dimensions.
- Plastic deformation leaves a permanent change after the force is removed.
- The elastic limit is the greatest stress a material can withstand without permanent deformation.

156. Explain how a force-extension graph gives the elastic potential energy stored in a material.

- Stretching an elastic material requires work, which is stored as elastic potential energy if the deformation is reversible.
- The stored energy equals the area under the loading force-extension graph.
- Within the limit of proportionality, the graph is straight, giving $U = \frac{1}{2}Fx = \frac{1}{2}kx^2$.